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Basic Information

Wang Rui (王睿)

Research Interest: CAR-T cell therapy; Spatial omics; Translational medicine; Imaging.
Main research: Identification and analysis of Spatially organized targets resistant to CAR-T cell therapy/immunotherapy across tumors

Skills

Linux, Python, R, MySQL; Cell Culture, MACSima; IELTS 6.5; Office/Markdown; AI Prompt design

Education

2022.9~2026.6	Zhejiang University	Stem cell and regenerative medicine	**Ph.D**
2017.9~2020.6	Nanjing Agricultural University	Crop genetics and breeding (bioinformatics)	**M.Sc.**
2013.9~2017.6	Nanjing Agricultural University	Agriculture	**B.Sc.**

Experience

2020.7~2022.6	R.A. Hangzhou First People's Hospital	Bioinformatics analysis; Lab management	**R.A.**

Researches

2022.9-Now	1. Explore spatial target of tumor microenvironment resistant to CAR-T cell therapy; 2. Computer-aided drug discovery and molecule dynamic simulation; 3. Other support.	**Ph.D**
2020.7-2022.6	1. Molecular subtyping for liver cancer; 2. Virtual screen for candidate targets; 3. Other necessary support.	**R.A.**
2017.9-2020.6	1. Construction and exploration of plant small RNA and their targets.	**M.Sc**

Publications and Fellowships

CAR-T cell therapy & spatial immune microenvironment (PhD)

1. **Rui Wang**#, …, Pengxu Qian*. [Working title: "CCR1-driven myeloid–fibroblast spatial barrier restricts CAR-T cell therapy in solid tumours" — manuscript under revision, first author]. *Cancer Cell*, 2026 (2nd minor revision / under revision). [First author (under revision)]

Highlight: Identified a CCR1-driven myeloid–fibrotic spatial immune barrier excluding CAR-T cells; pharmacologic disruption restored CAR-T infiltration and efficacy across tumor models. Patent filed (CCR1 antagonist). Keywords: Spatial omics / CAR-T resistance / target discovery / patent

2. Meng Zhu#, Yingli Han#, Tianning Gu#, **Rui Wang**#, …, He Huang, Pengxu Qian*. Class I HDAC inhibitors enhance antitumor efficacy and persistence of CAR-T cells by activation of the Wnt pathway. *Cell Reports*, 2024, 43(4): 114065. [Co-first author (4th)]

Highlight: Led multi-omics analysis in identifying HDACi-based epigenetic reprogramming of CAR-T cells. Keywords: Drug repurposing / epigenomics / CAR-T persistence

- Liver disease & transplant immunology multi-omics (Research Assistant)

3. Shuai Wang#, **Rui Wang**#, Nan Xu, Xuyong Wei, Yijie Yang, Zhengxing Lian, Beini Cen, Chenchen Shen, Wangyao Li, Jianguo Wang, Zhensheng Zhang, Linsong Tang, Qiang Wei, Di Lu, Xiao Xu*. SULT2B1-CS-DOCK2 axis regulates effector T-cell exhaustion in HCC microenvironment. *Hepatology*, 2023, 78(4): 1064–1078. [Co-first author (2nd)]

Highlight: Led multi-omics + immune-microenvironment analysis and virtual screening. Keywords: Tumor immunology / T-cell exhaustion / multi-omics/virtual screening

4. Di Lu#, Zuyuan Lin#, **Rui Wang**#, Zun Chen, Jianyong Zhuo, …, Shusen Zheng, Xiao Xu*. Multi-omics profiling reveals Chitinase-3-like protein 1 as a key mediator in the crosstalk between sarcopenia and liver cancer. *Redox Biology*, 2022, 58: 102538. [Co-first author (3rd)]

Highlight: Multi-omics integration identified CHI3L1 as a druggable mediator linking sarcopenia to liver cancer. Keywords: Inflammation-metabolism map/ biomarker discovery / druggable target/cytokine-metabolites omics

5. Xinyu Yang#, Di Lu#, **Rui Wang#**, Zhengxing Lian, Zuyuan Lin, …, Shusen Zheng, Xiao Xu*. Single-cell profiling reveals distinct immune phenotypes that contribute to ischaemia-reperfusion injury after steatotic liver transplantation. *Cell Proliferation*, 2021, 54(11): e13116. [Co-first author (3rd)]

Highlight: Single-cell dissection of immune phenotypes in transplant injury (clinical cohort data analysis). Keywords: Single-cell analysis / clinical cohort / transplant immunology

- Targeted delivery, nanomedicine & molecular simulation (collaboration)

6. Nasha Qiu#, Chang Xu#, Zhen Zhang#, **Rui Wang**#, Xuyong Wei#, …, Youqing Shen**, Xiao Xu*. Autologous tumoral esterase-driven therapeutic polymers sequentially orchestrated antigen-induction, STING activation and anti-angiogenesis for systemic cancer immune therapy. *Biomaterials*, 2025, 320: 123260. [Co-first author (4th)]

Highlight: Conducted comparative molecular docking of polymer-derived metabolites and supported downstream RNA-seq analysis. Keywords: Nanomedicine / drug delivery / molecular docking

7. Qiwei Wang#, Jingyi Li#, Xin Zeng#, Lina Liu#, Huan Lu#, …, **Rui Wang**, …, Shanshan Pei, Pengxu Qian*. TIM3-targeted delivery of venetoclax overcomes drug resistance and reinvigorates NK cell activity in acute myeloid leukemia. *Nature Cancer*, 2026 (online). [Co-author (8th)]

Highlight: Contributed to all-atoms molecular dynamics simulations characterizing the self-assembly of venetoclax and IR-783-NHS into functionalized nanoparticles. Keywords: Targeted delivery / drug resistance / molecular dynamics

- Plant small RNA & degradome bioinformatics (MSc)

8. **Rui Wang**, Zhongyi Yang, Yuhan Fei, Jiejie Feng, Hui Zhu, Fang Huang, Hongsheng Zhang, Ji Huang*. Construction and analysis of degradome-dependent microRNA regulatory networks in soybean. *BMC Genomics*, 2019, 20: 465. [First author]
Highlight: Built degradome-based miRNA–target regulatory networks in soybean; established the bioinformatic foundation that later underpinned target-discovery pipelines. Keywords: Small RNA / degradome / network bioinformatics

9. Yuhan Fei, **Rui Wang**, Haoyuan Li, Shu Liu, Hongsheng Zhang, Ji Huang*. DPMIND: degradome-based plant miRNA–target interaction and network database. *Bioinformatics*, 2018, 34(9): 1618–1620. [Second author]

Highlight: Co-developed a public plant miRNA–target database and web platform (data engineering + web service delivery). Keywords: Database / web platform / reproducible pipeline

Patent: Application of CCR1 antagonists in enhancing CAR-T cell antitumor efficacy (ID: 02511662659.8, under substantive examination)

Full publication record: Google Scholar — https://scholar.google.com/citations?user=umG7RH4AAAAJ&hl=en

Funding

2022.1~2024.12	Provincial Clinical Technology Application Project of the Health Commission of Zhejiang Province (Completed)	**Project Lead**

Conference

2025.5	10th Chinese Congress of Hematology	**Poster Presenter**
2024-Present	Chinese Association for Physiological Sciences (CAPS); Chinese Society of Biotechnology	**Student Member**
2026.12	ASH 2026, U.S, New Orleans	**Poster Presenter**

Awards

2025.10	National Scholarship for Doctoral Students	**2018~2019**	1. National Scholarship for Master's Students (awarded twice)
2025.10	Outstanding Graduate Student, PhD University		2. Master's Academic Scholarship (multiple awards)
2025.1	Selected for the Doctoral Special Program of the CAST Youth Talent Promotion Program		